

Movie Recommendation System Using Collaborative Filtering

Kónya Tamás Ádám
konyatomi16@stud.uni-obuda.hu
B6EHWG

Abstract - Recommender systems play a key role in modern online services by helping users find relevant content within large datasets. The aim of this research is to investigate the performance of movie recommendation systems based on collaborative filtering. The study compares user-based and item-based collaborative filtering methods using the MovieLens (ml-latest-small) dataset, which contains more than 100,000 ratings. The system was implemented in Python, with similarity calculated using Pearson correlation and predictions generated using a neighborhood-based method. Model performance was evaluated in terms of prediction accuracy (RMSE, MAE) and recommendation quality (Precision@10, Recall@10) under different train-test splits. The results show that, in these experiments, the user-based method performed better than the item-based approach across all evaluated metrics. The 75/25 train-test split provided the best prediction accuracy for the user-based method (RMSE = 0.970; MAE = 0.749), whereas the 80/20 split resulted in the highest Recall@10. These findings indicate that the performance of collaborative filtering depends not only on the selected method but also on the train-test ratio and the evaluation objective.

Keywords - recommender system, collaborative filtering, user-based filtering, item-based filtering, MovieLens, machine learning, RMSE, MAE, Precision@k, Recall@k

I. INTRODUCTION

Recommender systems play an increasingly important role in online services, helping users identify content that is relevant to them within the vast amount of available information. One of the most commonly used and well-established approaches in movie recommendation systems is collaborative filtering. This method relies on users' behavioral data and ratings to generate recommendations. Its fundamental assumption is that users with similar preferences are likely to enjoy similar movies in the future as well.

Although collaborative filtering is widely used, research has also identified certain limitations of the method, such as data sparsity and the so-called cold-start problem. These factors can significantly reduce recommendation accuracy, particularly when new users appear or new content is added. Although the literature proposes several advanced approaches to mitigate these issues, including hybrid systems [2], deep reinforcement learning [3], and K-means clustering [4], the appropriate selection and optimization of fundamental collaborative filtering techniques remain essential for building effective systems.

The main objective of this research is to conduct a comprehensive examination of a movie recommendation system by applying and comparing different collaborative filtering methods. The experiment specifically compares the performance of user-based and item-based collaborative filtering techniques. The system uses the MovieLens (ml-latest-small) dataset, which contains more than one hundred thousand ratings, and the results are analyzed using rigorous

metrics: RMSE and MAE for prediction accuracy, and Precision@k and Recall@k for recommendation quality.

The study is structured as follows: Section II (Literature Review) reviews the relevant literature related to recommender systems and collaborative filtering, as well as previous research directions in the field. Section III (Methodology) describes the dataset, the data-processing workflow, the algorithms examined, and the evaluation metrics in detail. Section IV (Results) presents the quantitative results of the experiments and compares the performance of the methods under investigation. Finally, Section V (Conclusion) summarizes the main findings of the research, interprets their significance, and discusses the limitations of the study and possible directions for future research.

II. LITERATURE REVIEW

Recommender systems play an important role in modern online services by helping users find relevant content among large volumes of information. One of the most commonly used approaches in movie recommendation systems is collaborative filtering, which generates recommendations based on users' behavioral data and ratings. The method is based on the assumption that users with similar preferences are likely to enjoy similar movies.

Research shows that movie recommendation systems use different approaches to model user preferences. The review study by Goyani and Chaurasiya demonstrates that recommender systems are often based on collaborative filtering or content-based methods, or on a combination of the two to achieve more effective recommendations [1]. Collaborative filtering is especially common on platforms that contain large amounts of user-rating data.

However, some studies also point out the limitations of collaborative filtering, including the so-called cold-start problem and data sparsity. These issues can reduce recommendation accuracy, particularly for new users or newly added content. Widayanti et al. propose a hybrid approach that combines collaborative filtering and content-based filtering in order to improve recommender-system performance and increase recommendation accuracy [2]. Such methods can help improve the effectiveness of recommender systems while reducing the limitations of traditional approaches.

The typical challenges of recommender systems, including the cold-start problem and data sparsity, were already analyzed in detail by Adomavicius and Tuzhilin [10] in their seminal work. Their study provides a comprehensive overview of the effects of collaborative and content-based filtering and laid the early theoretical foundations for hybrid approaches and context-aware recommendations, which continue to shape research directions today.

Further advanced solutions have been developed to address the shortcomings of traditional collaborative filtering,

such as data sparsity and the cold-start problem. Peng et al. [3] propose an innovative system that integrates deep reinforcement learning (DRL) with collaborative filtering. Their approach applies the "actor-critic" method and the DDPG algorithm to generate initial recommendations for new users, while singular value decomposition (SVD) is used during matrix factorization to extract more detailed features. Another direction is represented by the research of Phorasim and Yu [4], who combined collaborative filtering with the K-means clustering algorithm. Their method groups users in multidimensional spaces according to their interests, which not only increases accuracy compared with traditional collaborative filtering but also significantly reduces computation time.

Among the classical neighborhood-based approaches to collaborative filtering, the item-based technique introduced by Sarwar et al. [8] is particularly important alongside user-based methods. Instead of focusing on similarities between users, this approach considers similarities between items (movies), offering substantial scalability and computational advantages for large databases. As a result, it is often used as a robust baseline model in comparative analyses.

The development of collaborative filtering continued in two directions. Dev and Mohan [5] focused on scalability by distributing the workload across multiple computers, allowing collaborative filtering to remain fast even for large databases [5].

Katarya and Verma [6], by contrast, focused on improving accuracy. They combined three techniques: asymmetric similarity computation, matrix factorization, and typicality-based filtering. Asymmetric similarity recognizes that similarity is not necessarily mutual, which can more accurately reflect real user relationships [6]. Matrix factorization reveals hidden patterns in the data, while the typicality-based method groups movies by genre, improving prediction accuracy particularly when the data are sparse [6].

In the extraction of latent features, model-based collaborative filtering, particularly matrix factorization, represented a major breakthrough. As demonstrated by Koren et al. [9], mapping user-item interactions into a lower-dimensional space can handle data sparsity very effectively and produce fine-grained, accurate recommendations by identifying latent preferences.

While Dev and Mohan [5] focused on speed and the handling of large datasets, Katarya and Verma [6] concentrated on more accurate predictions. Combining the two approaches may be a promising direction for future research, since a system that is both fast and accurate could be an ideal choice for real-world applications.

III. METHODOLOGY

The aim of the research is to examine a movie recommendation system using different collaborative filtering methods. The system generates recommendations based on users' previous movie ratings, after which the quality of the recommendations is analyzed using different evaluation metrics.

The MovieLens dataset, specifically the ml-latest-small version, is used for the experiment. The dataset contains more than one hundred thousand movie ratings describing interactions between users and movies. The data include user

IDs, movie IDs, and rating scores, from which the user-movie rating matrix can be constructed.

The system is implemented in the Python programming language. The Pandas library is used for data processing, allowing CSV files to be loaded and managed in a DataFrame structure and enabling the user-movie rating matrix to be created using a pivot table. NumPy is used for numerical calculations and for determining similarity values. To examine the effect of the train-test ratio on model performance, the dataset was evaluated using three different splits: 70/30, 75/25, and 80/20. The training and test sets were generated using the `train_test_split` function of the Scikit-learn library with a fixed random state of 42 to ensure reproducibility. During evaluation, only test instances for which both the user and the movie were present in the corresponding training set were considered, thereby excluding cold-start cases from the reported metrics.

Two collaborative filtering methods are compared in the research: user-based and item-based collaborative filtering. In both methods, similarity is calculated using Pearson correlation. A minimum of two common ratings is required for the similarity calculation, and only positive similarity values are considered during prediction. The models use up to 30 nearest neighbors. In the user-based approach, the ratings of the most similar users who rated the target movie are combined using a similarity-weighted average. In the item-based approach, the prediction is based on the most similar movies previously rated by the target user, also using a similarity-weighted average.

The performance of the system is evaluated from two different perspectives. Prediction accuracy is measured using RMSE (Root Mean Squared Error) and MAE (Mean Absolute Error), which measure the differences between predicted and actual ratings in the test set. Recommendation quality is evaluated using Precision@10 and Recall@10. For this evaluation, a movie is considered relevant if the user's actual rating in the test set is 4.0 or higher. For each user, movies in the test set are ranked according to their predicted ratings, and the ten highest-ranked movies are treated as recommendations. Precision@10 measures the proportion of relevant movies among these top 10 recommendations, while Recall@10 measures the proportion of all relevant movies in the user's test set that appear among the top 10 recommendations. The final Precision@10 and Recall@10 values are obtained by averaging the user-level results.

The scope of the research is defined by certain limitations. The study does not address content-based recommender systems, does not apply deep learning models, and does not aim to develop a complete recommender-system platform. In addition, the research does not include user experiments; the evaluation is based exclusively on the results of experiments conducted on the available dataset.

IV. RESULTS

The experiments compare two approaches to collaborative filtering, user-based and item-based methods, under different train-test splits. The performance of the system was evaluated using RMSE (Root Mean Squared Error) and MAE (Mean Absolute Error) for prediction accuracy, and Precision@10 and Recall@10 for recommendation quality.

The prediction-accuracy results are shown in Table I.

TABLE I PREDICTION ACCURACY (RMSE AND MAE)

Train-test split	Method	RMSE	MAE
70/30	User-based	0.978530	0.754748
70/30	Item-based	0.984761	0.761577
75/25	User-based	0.970426	0.749167
75/25	Item-based	0.980674	0.759564
80/20	User-based	0.972262	0.750189
80/20	Item-based	0.983170	0.762194

It can be observed that for every split examined, the user-based method achieves lower RMSE and MAE values than the item-based approach. The best result is obtained by the user-based method with the 75/25 split (RMSE = 0.970426, MAE = 0.749167), indicating that this configuration provides the most accurate rating predictions. The item-based method produces higher errors in every case, indicating weaker prediction performance.

The quality of the recommendations based on the Precision@10 and Recall@10 metrics is shown in Table II.

TABLE II RECOMMENDATION QUALITY (PRECISION@10 AND RECALL@10)

Train-test split	Method	Precision@10	Recall@10
70/30	User-based	0.618033	0.581367
70/30	Item-based	0.539836	0.545025
75/25	User-based	0.592131	0.622212
75/25	Item-based	0.520000	0.584486
80/20	User-based	0.555410	0.674679
80/20	Item-based	0.491475	0.639732

Based on the Precision@10 values, the user-based method performs better for every split, showing that a higher proportion of relevant items appears among the top recommendations. For Recall@10, it can be observed that the larger training set (80/20 split) results in higher recall values, particularly for the user-based method (Recall@10 = 0.674679). This suggests that with more training data, the system is able to identify a larger number of relevant items.

In the study, item-based collaborative filtering was used as the baseline method against which the user-based approach was evaluated. In these experiments, the user-based method performed better than the item-based baseline across all reported metrics, both in terms of prediction accuracy and recommendation quality. These results suggest that, for the examined dataset and experimental setup, modeling similarities between users captured preference relationships more effectively than modeling similarities between items.

Overall, user-based collaborative filtering performed better than the item-based method in these experiments, particularly in terms of lower prediction errors and a higher proportion of relevant recommendations. The 75/25 split provided the best prediction accuracy, whereas the 80/20 split

resulted in the highest Recall@10. Therefore, the preferred train-test split depends on the evaluation objective.

To ensure full reproducibility of the experiments, the implementation and all experiments performed are available on the Kaggle platform [7]. The repository contains the dataset used, the implementation of the models, and all code and outputs used to generate the results.

V. CONCLUSION

The aim of this research was to examine a movie recommendation system using different collaborative filtering methods, with particular emphasis on comparing user-based and item-based approaches. The results show that, in these experiments, user-based collaborative filtering performed better across all examined metrics. It achieved lower RMSE and MAE values as well as higher Precision@10 and Recall@10 scores, corresponding to more accurate predictions and more relevant recommendations. The best prediction result was achieved with the 75/25 train-test split, while the larger training set (80/20) contributed to higher recall values.

These results suggest that, within the examined dataset and experimental setup, modeling similarities between users captured relationships between preferences more effectively than modeling similarities between items. The experiments also show that the train-test ratio influenced the evaluated performance metrics: the 75/25 split produced the lowest RMSE and MAE values, whereas the 80/20 split achieved the highest Recall@10. Therefore, the selection of the train-test ratio should depend on whether prediction accuracy or the retrieval of relevant items is prioritized.

The research nevertheless has certain limitations. The study was limited exclusively to traditional collaborative filtering methods, did not include hybrid or deep-learning-based approaches, and did not examine the effect of real user feedback. In addition, the evaluation was based on a single dataset, which may limit the generalizability of the results.

Future research could investigate the application of hybrid recommender systems that combine collaborative and content-based approaches, as well as consider incorporating modern machine learning techniques such as matrix factorization or deep neural networks. Further directions could include validation on multiple datasets and testing in real user environments. These developments could contribute to the creation of an even more accurate and robust recommender system.

REFERENCES

- [1] M. Goyani and N. Chaurasiya, "A review of movie recommendation system: Limitations, survey and challenges," *ELCVIA: Electronic Letters on Computer Vision and Image Analysis*, vol. 19, no. 3, pp. 18–37, 2020.
- [2] R. Widayanti, M. H. R. Chakim, C. Lukita, U. Rahardja, and N. Lutfiani, "Improving Recommender Systems using Hybrid Techniques of Collaborative Filtering and Content-Based Filtering," *Journal of Applied Data Sciences*, vol. 4, no. 3, pp. 289–302, 2023.
- [3] S. Peng, S. Siet, S. Ilkhomjon, D.-Y. Kim, and D.-S. Park, "Integration of Deep Reinforcement Learning with Collaborative Filtering for Movie Recommendation Systems," *Appl. Sci.*, vol. 14, no. 3, p. 1155, Jan. 2024, doi: 10.3390/app14031155.
- [4] P. Phorasim and L. Yu, "Movies recommendation system using collaborative filtering and k-means," *Int. J. Adv. Comput. Res.*, vol. 7, no. 29, pp. 52–59, 2017, doi: 10.19101/IJACR.2017.729004.

- [5] A. V. Dev and A. Mohan, "Recommendation system for big data applications based on set similarity of user preferences," in 2016 International Conference on Next Generation Intelligent Systems (ICNGIS), Kottayam, 2016, pp. 1-6, doi: 10.1109/ICNGIS.2016.7854058.
- [6] R. Katarya and O. P. Verma, "Effective collaborative movie recommender system using asymmetric user similarity and matrix factorization," in 2016 International Conference on Computing, Communication and Automation (ICCCA), Noida, 2016, pp. 71-75, doi: 10.1109/CCAA.2016.7813692.
- [7] T. Kónya, "Movie Recommendation System in Kaggle," Kaggle Notebook, 2025. [Online]. Available: <https://www.kaggle.com/code/tamsknya/movierecommendationsystem-in-kaggle>
- [8] B. Sarwar, G. Karypis, J. Konstan, and J. Riedl, "Item-based collaborative filtering recommendation algorithms," in Proceedings of the 10th international conference on World Wide Web (WWW '01), 2001, pp. 285-295, doi: 10.1145/371920.372071.
- [9] Y. Koren, R. Bell, and C. Volinsky, "Matrix Factorization Techniques for Recommender Systems," *Computer*, vol. 42, no. 8, pp. 30-37, Aug. 2009, doi: 10.1109/MC.2009.263.
- [10] G. Adomavicius and A. Tuzhilin, "Toward the next generation of recommender systems: a survey of the state-of-the-art and possible extensions," *IEEE Transactions on Knowledge and Data Engineering*, vol. 17, no. 6, pp. 734-749, June 2005, doi: 10.1109/TKDE.2005.99